study and comparison with those not providing such therapy. We feel that attention must be paid simultaneously to the intrapsychic problems and the external community and home problems in young children of the type studied here.

REFERENCES

1. Stark W, Bentzen F: The integration of the emotionally disturbed child in the normal school setting, in Orthopsychiatry and the School. Edited by Krugman M. New York, American Orthopsychiatric Association, 1958, pp 82-95
2. Riese H: Educational therapy, a methodological approach to the problem of the untreatable child. Group Psychotherapy 12:58-66, 1959
3. Leve RM, Graffagnino, PN, Avallone SF: An attempt to combine clinical and educational resources: report of the first year's experience of a therapeutic school. J Amer Acad Child Psychiat (in press)
4. Leve RM, Fitzgibbons D, Breetz C: An evaluation of effects of special education and psychotherapy on emotionally disturbed children in an inner-city school. Presented at the annual meeting of the American Association of Psychiatric Clinics for Children, Boston, Mass, Nov 2-3, 1969
5. Graffagnino PN, Boelhouwer C, Reznikoff M: An organic factor in patients of a child psychiatric clinic: data from the early interviews and the electroencephalogram. J Amer Acad Child Psychiat 7:618-638, 1968
6. Hollingshead AB, Redlich FC: Social Class and Mental Illness. New York, John Wiley & Sons, 1958
7. Freud A: Assessment of childhood disturbances, in Psychoanalytic Study of the Child, vol 17. Edited by Eissler RS, Freud A, Glover E, et al. New York, International Universities Press, 1962, pp 149-158
8. Wolfenstein M: Loss, rage and repetition, in The Psychoanalytic Study of the Child, vol 24. Edited by Eissler RS, Freud A, Glover E, et al. New York, International Universities Press, 1969, pp 432-460
9. van Amerongen ST, Hiltner R, Huntington D: Children and Clinics: A Survey. New York, American Association of Psychiatric Clinics for Children, 1968, pp 3-6
10. Harrison SI, McDermott JF, Wilson PT, et al: Social class and mental illness in children: choice of treatment. Arch Gen Psychiat 13:411-417, 1965

Brain Function in Problem Children and Controls: Psychometric, Neurological, and Electroencephalographic Comparisons

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Psychometric, neurological, and EEG studies were made of 24 children with scholastic-behavioral problems but no classical evidence of neurological disease and of 24 matched controls. Differences between the two groups were significant in all three measures, giving evidence of brain dysfunction in the groups with scholastic-behavioral problems. Two subgroups—hyperactive and nonhyperactive—each with characteristics suggesting a different syndrome, are described.

SOME 30 YEARS AGO, Strauss and Kephart (1) and Strauss and Werner (2) differentiated two types of mentally defective children residing in a training school. In contrast to an "endogenous" group, an "exogenous" group manifested "erratic, uncoordinated, uncontrolled and socially unacceptable" behavior (1), and on the Marble Board test, showed impairment of ability to reproduce

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geometric designs and domination by the background, when it was prominent (2). Although only three to five percent of the exogenous group showed evidence of a gross neurological disturbance, this group was considered to be "brain injured" on the basis of history and (unspecified) neurological signs.

Since then, the concept of "minimal brain damage" (or, more guardedly, "minimal brain dysfunction") has been extended to children who may have normal or superior intelligence but manifest several, if not all, of the following: hyperactivity, perceptual-motor impairments, emotional lability, general coordination deficits, disorders of attention, impulsivity, disorders of memory and thinking, specific learning disabilities, disorders of speech and hearing, equivocal neurological signs, and EEG abnormalities (3).

In the absence of correlative data with direct evidence of "brain damage" (e.g., neuroanatomical and neurochemical changes at autopsy or biopsy), it is difficult to say which, if any, of these signs are primary indicators of brain damage and which are secondary consequences. Nevertheless, a survey of the more recent literature (4-13) reveals that EEG abnormalities of various sorts (mainly, slow activity in excess of age-expectancy) are very common among children with behavior disorders, particularly those designated as "hyperactive" or "hyperkinetic." Also, a few studies (14-16) indicate that equivocal or "soft" neurological signs are more common among children with behavior disorders than in control subjects, even in the absence of classical or "hard" signs of neurological disease in the former. Likewise, perceptual and other psychological deficits have been found frequently in children with behavior disorders (11, 17-19).

However, except in the EEG study of Laufer and associates (6) and in the neurological study of Hertzig and associates (16), no attempt has been made to compare children exhibiting hyperactive and nonhyperactive behavior disorders with each other, and no reports have come to our attention that compare hyperactive and nonhyperactive behavior problem children with matched controls in respect to all three presumptive indicators of brain damage.

The purposes of the present investigations were to delineate, in quantitative terms, the psychometric, neurological, and EEG characteristics that differentiate children with scholastic-behavioral disorders (so designated because scholastic and behavioral problems were inextricably mixed) *but without classical signs of neurological disease* from control children matched for sex, race, age, intelligence, and socioeconomic class, and to determine whether in the former, hyperactive and nonhyperactive children differ from each other in respect to these characteristics. While the results of such a study could not be expected to furnish unequivocal proof of the existence or nonexistence of brain damage in a population of this sort, it was hoped that the investigations would help define the kinds of brain dysfunction that may be found in such patients and the relationships among such dysfunctions.

Methods

Subjects

A group of 25 children aged five to 15 years referred to the outpatient clinic of the department of psychiatry, University of Kentucky, for scholastic and/or behavior problems were selected as subjects for the study.

Twenty-two of the subjects were referred by teachers and/or parents and three by physicians. Referrals for scholastic problems alone (underachievement, with reading problems in most cases) and for hyperactivity or hyperactivity and aggressiveness were represented about equally. Only two children were described as "withdrawn" and "resistant."

The following children were excluded: those who showed evidence of organic neurological disease on the initial screening examination, those with clinical seizures, those with IQs below 86, and those on chemotherapy (except amphetamine drugs). Children receiving amphetamine drugs on admission were tested and examined both on the drug and later, several weeks after drug withdrawal. Included in this report are only the test results and examination findings under drug-free conditions. One child, later

found to have definite signs of cerebral palsy, was eliminated from the patient group, leaving 24 subjects (23 Caucasian and one Negro; 22 boys and two girls).

A group of matched control subjects was obtained through the cooperation of the Fayette County schools. Matching was on the basis of sex, age, race, intelligence, general social class, and the absence of any indication from school records and school personnel that the child had demonstrated scholastic or behavioral problems during his school years. The child matched with the one showing signs of cerebral palsy was also later omitted, leaving 24 control subjects.

Procedure

Children referred to the clinic formed the original pool of patients from which the final 24 subjects were chosen according to the criteria mentioned above. Selection of patient subjects as suitable for inclusion was made at various points throughout the diagnostic evaluation, which included a detailed social service report and preliminary psychiatric and neurological screening examinations. When the selection criteria were met, parental permission was obtained for the child's inclusion in the study. The selection and matching of the control subjects were done initially by consulting the school records, with the assistance of the principal and teachers; these records contained most of the necessary information, including results of previous intelligence and achievement tests.

The entire evaluation of the patient subjects, including special procedures, usually extended over a two-to-three-week period. However, the various studies done with the control subjects were ordinarily completed in one day. The principal usually obtained the permission of the parents for their child's participation in the study; if the parents were agreeable, the child was picked up at school in the morning, brought to the medical center for the day, and returned to his home in the afternoon.

Psychometric Testing

A group of tests that could be objectively scored was administered to each child. These tests included the Wechsler Intelligence Test for Children, (WISC), the Bender Motor Gestalt Test, the Graham-Kendall Memory for Designs Test, the Minnesota Percepto-Diagnostic Test, the Draw-a-Person Test, and the Wide Range Achievement Test. In the case of the control subjects, the results of the previous intelligence tests at the school were compared with the results obtained in the psychometric evaluation before the child was finally included. Occasionally a child patient did not perform on a particular test, for example because of refusal, in which case the results for his control were also omitted from the statistical analysis.

Neurological Examination

In addition to a complete classical neurological examination, each child was tested for the presence of soft signs culled from the literature. Other deviations from classical standards of neurological normality were also recorded. In the case of any individual soft sign, a single notation of "present" was made if certain criteria (necessarily subjective) were met, as illustrated by the following for those signs that differentiated the patient from the control group to a statistically significant degree:

1. *Equivocal Babinski*, etc. Positive responses to the Babinski test or any of its confirmatories in not fewer than two or more than three of five trials.

2. *Dysdiadochokinesis.* Performance poorer than expected for age on rapid alternation and pronation of the hands and/or rapid apposition of the thumb to each of the other four digits, persisting on repeated tests with variations in instructions (to assure comprehension).

3. *Lateral nystagmoid jerks.* Unsustained, irregular, oscillatory movements of the eyes on lateral or upward gaze in at least three of five trials.

4. *Unilateral winking defect.* Ability to wink with one eye but not with the other in at least four of five trials.

5. *Overflow movements.* Adventitious convergence and divergence or choreiform movements of the fingers, athetoid movements of the arms or repeated opening and closing of the eyes, standing in the Romberg position, eyes closed, upper extremities extended horizontally and forward, fingers separated and extended, counting aloud to 25, on repeated tests with admonitions to

refrain from such movements.

6. *Disturbances of gait and/or coordination and/or muscle tone.* On at least two tests, loss of associated arm swings (one or both sides), awkwardness in running or catching a ball, inability to hop on one foot repetitively within a circumscribed area, swaying in Romberg position (eyes closed), generalized hypotonia or hypertonia (slight cogwheel rigidity)—singly or combined.

The neurological examinations were made in accordance with a form prepared, and instructions and demonstrations given, by the senior author (A.W.). Some of the subjects in both the patient and control groups were examined by the senior author, while others were examined by various members of the department of psychiatry (see Acknowledgments). In most cases where classical or soft signs were observed by examiners other than the senior author, they were checked by the latter as soon as possible after the complete neurological examination. Notations of "present" on the tabulation sheets were made only if the check confirmed the original observation; in most instances, this was actually the case.

TABLE 1
Profile of the Hyperactive Child

I.	AT HOME
	1. Cannot remain still
	2. Cannot conform to limits or prohibitions
	3. Makes excessive demands
	4. Has sleeping problems
	5. Shows unwarranted aggression
	6. Is general "pest"
II.	AT SCHOOL
	1. Is talkative
	2. Fidgets continuously
	3. Cannot concentrate
	4. Has short attention span
	5. Cannot conform to limits or prohibitions
	6. Shows poor school achievement
III.	RELATIONSHIPS WITH OTHER CHILDREN
	1. Cannot make friends
	2. Fights without provocation
	3. Has poor manners
	4. Is extremely bossy
	5. Disregards rights of others
	6. Is constantly rejected

* Reproduced with the permission of Parke, Davis and Company, Detroit, Mich., from Therapeutic Notes 68: 176–180 (June) 1961

Electroencephalograms

A Grass Model IV eight-channel electroencephalogram was used to obtain bipolar and monopolar (ear reference) tracings from eight disc electrode placements on each side of the head (international system), with the child semirecumbent in an adjustable upholstered chair, eyes closed, awake but as relaxed as possible. The first sequence of runs were, on each side: bipolar, serially linked parasagittal; the same during two minutes hyperventilation, continued until complete recovery; and bipolar, serially linked prefrontal-anterior temporal-mid-temporal-posterior temporal-occipital. Subsequently other bipolar runs, and finally monopolar runs (referred to the homolateral and to the contralateral ears), were made. No medications were given, but after the first (waking) sequence of runs, spontaneous drowsy or sleep records were obtained in 18 of the patients and 13 of the control subjects.

Evaluation of the electroencephalograms was made in two ways:

1. The tracing from each channel on the first and third runs (waking) was analyzed quantitatively by visual-manual measurement of the percent time of artifact-free record (at least two minutes) occupied by *rhythmic* activity (defined as three or more successive waves of the same frequency ±0.5 c.p.s.), in the beta (14.0-25.0 c.p.s.), alpha (8.0-13.0 c.p.s.), slow alpha (7.0-7.9 c.p.s.), theta (4.0-6.9 c.p.s.), and delta (less than 4.0 c.p.s.) ranges. Tabulation of the data and computation of means revealed that the most striking difference between the patient and control subjects was in the amount of slow alpha and theta rhythms (very little delta activity was found in the records of either the patients or the controls). As these rhythms, when present, were quite diffuse, a single index of slow activity was obtained for each child, namely "mean total slow percent" ($\overline{TS}\%$), computed by dividing the sum of percent time the tracings from each channel were occupied by 4.0-7.9 c.p.s. rhythmic activity, by the number of channels. Generally, the number of channels was 16 (eight each in the first and third runs); in some cases, however, the computations for both the numerator and denominator were based on fewer channels because 16 artifact-free tracings of two-minute duration (each run) could not be obtained.

2. All tracings were inspected for gross bilateral asymmetries, paroxysmal discharges, spikes or spike-wave complexes, "14 and 6" rhythms, focal slow activity, prolonged slowing after hyperventilation,

TABLE 2

Differences Between Patients and Matched Control Subjects on Psychometric Tests (t Test for Correlated Groups)

TEST	N (PAIRS)	MEAN SCORE (PATIENTS)	MEAN SCORE (CONTROLS)	MEAN DIFFERENCE	S.D.	t	p<
Wechsler Intelligence Test for Children (WISC)							
Full scale IQ	23	110.13	111.17	-1.04	10.86	0.46	NS
Verbal IQ (V)	23	112.65	113.22	-0.57	13.35	0.22	NS
Performance IQ (P)	23	105.57	106.91	-1.34	16.11	0.40	NS
Difference V-P (patients)		112.61(v)					
	23	105.57(p)		7.04	13.61	2.48	.05
Difference V-P (controls)			113.22(v)				
	23		106.91(p)	6.31	16.53	1.83	NS
Coding subtest*	23	10.22	11.78	-1.56	3.45	2.18	.05
Draw-a-Person IQ	22	94.36	96.23	-1.87	27.62	0.32	NS
Bender-Gestalt (Koppitz) (errors)	23	2.83	0.39	2.44	3.49	3.35	.01
Graham-Kendall (errors)	23	4.48	1.35	3.13	7.18	2.09	.05
Minnesota Percepto-Diagnostic (errors)	21	49.48	37.38	12.10	17.92	3.09	.01
Wide Range Achievement							
Reading grade level	24	5.54	5.30	0.24	2.52	0.46	NS
Spelling grade level	24	4.48	4.56	-0.08	1.98	0.21	NS
Arithmetic grade level	24	4.61	4.17	0.44	1.58	1.37	NS

*None of the other WISC subtests showed significant differences between patients and controls.

and other pathological features. These were recorded as present or absent in each case.

Hyperactivity

Children in the-patient group (only) were classified "hyperactive," "nonhyperactive," or "uncertain" on the basis of a check list (see table 1) completed by the social worker in consultation with the parents and notations made independently by the participating psychiatrists during the intake interview with the patient and parent(s). A child was classified "hyperactive" if at least 12 of the 18 items on the check list were affirmed and if "hyperactivity" was mentioned in the psychiatrist's report. If fewer than 12 of 18 items were affirmed on the check list and the psychiatrist did not mention "hyperactivity," the child was classified "nonhyperactive." The classification "uncertain" consisted of those children for whom the check list and psychiatrist's notation did not agree or for whom no check list was completed. Using these criteria, 11 of the children were classified as hyperactive and nine nonhyperactive.

Statistics

For analysis of measures on interval scales (psychometric scores, ages, $\overline{TS}$%), the t test, analysis of variance and covariance, or product-moment correlation coefficient was employed when appropriate (20, 21). For measures on nominal scales the sign test or chi square and on ordinal scales the Wilcoxon matched-pairs signed-ranks test, the Mann-Whitney U test or the Spearman rank correlation coefficient were used where indicated (22).

Results

Psychometric Scores

The following tests (see table 2) differentiated the patient children from their matched controls to a statistically significant degree: difference between the verbal and performance subtests of the WISC; the coding subtest of the WISC; and the Bender-Gestalt, Graham-Kendall, and Minnesota Percepto-Diagnostic. On all of these tests the patient children's performance was poorer than that of their controls. As might be expected because of the matching procedure, no significant difference was found on the WISC Full Scale I.Q. Not expected, however, was the finding that the patients and controls did not differ on the Wide-Range Achievement Test.

On the other hand, comparison of the hyperactive with the nonhyperactive patient children (see table 3) reveals the significantly poorer performance of the hyper-

active children not only on the Bender-Gestalt and Graham-Kendall tests but also on the Wide-Range Achievement Test. In this connection, it should be noted that the mean age of the hyperactive group was 2.8 years less than that of the nonhyperactive group.

Comparison of the hyperactive and nonhyperactive children with their respective controls (see table 4) reveals that whereas the hyperactive group had significantly lower scores than their controls on the WISC Full Scale I.Q., Performance I.Q., Bender-Gestalt, Graham-Kendall, and WRA reading and spelling tests, this was not true of the nonhyperactive group. Indeed, comparison of the latter with their controls reveals superior scores on the part of the nonhyperactive patient group on several of the psychometric tests, namely the WISC Full Scale I.Q., Performance I.Q., and the Draw-A-Person and WRA reading and arithmetic tests.

Neurological Signs

After exclusion of the patient child with cerebral palsy (and her matched control), none of the 24 patient children was found to have any positive classical signs; one of the matched control subjects showed bilateral absence of associated arm swing on walking and a unilateral positive Babinski sign (etiology not apparent). Twenty-two of the 24 patients and 20 of the 24 matched control children were found to have one or more soft neurological signs; in all there were 92 soft signs in the patient group and 39 in the control group. This difference is significant (see table 5). On the other hand, the hyperactive and nonhyperactive patient subgroups as well as their respective control subgroups did not differ significantly from each other in respect to total number of soft signs, although the number of such signs was significantly greater in each of the patient subgroups than in their respective control subjects (table 4).

Of the individual soft signs computed separately (all others disregarded), Equivocal Babinski, etc., and Dysdiadochokinesis were significantly more common in the patient group by a two-tailed test, and two additional ones (Lateral Nystagmoid Jerks and Unilateral Winking Defect) by a one-tailed test. Curiously, Overflow Movements were more common in the matched control group; the difference was significant by a one-tailed test. Combining one or more soft signs under the heading Disturbance of Gait and/or Coordination and/or Muscle Tone, a significantly larger number of patient children displayed what might be called global clumsiness than did controls (see table 6). Other less common soft signs that did not differentiate the patient from the control children significantly were: asym-

TABLE 3

Differences Between Hyperactive and Nonhyperactive Patients in Psychometric Tests (t Test for Independent Groups)

TEST	HYPERACTIVE PATIENTS N	HYPERACTIVE PATIENTS MEAN SCORE	HYPERACTIVE PATIENTS S.D.	NONHYPERACTIVE PATIENTS N	NONHYPERACTIVE PATIENTS MEAN SCORE	NONHYPERACTIVE PATIENTS S.D.	MEAN DIFFERENCE	t	p<
Wechsler Intelligence Test for Children (WISC)									
Full Scale IQ	11	105.73	24.70	8	115.00	15.63	-9.27	0.93	NS
Verbal IQ (V)	11	108.00	24.17	8	117.50	15.90	-9.50	0.97	NS
Performance IQ (P)	11	102.36	22.45	8	109.25	16.58	-6.89	0.73	NS
Difference (V-P)	11	6.30	14.85	8	8.25	15.21	-1.95	0.30	NS
Draw-a-Person IQ	10	90.00	21.39	8	94.75	20.16	-4.75	0.48	NS
Bender-Gestalt (Koppitz) (errors)	11	4.82	5.02	8	0.75	1.16	4.07	2.23	.05
Graham-Kendall (errors)	10	7.80	9.33	8	1.12	1.12	6.68	2.00	.05*
Minnesota Percepto-Diagnostic (errors)	10	56.90	31.80	8	44.37	15.41	12.53	1.02	NS
Wide Range Achievement									
Reading grade level	11	3.37	2.18	9	8.34	4.29	-4.97	3.36	.01
Spelling grade level	11	2.81	1.67	9	6.52	3.62	-3.71	3.04	.01
Arithmetic grade level	11	3.01	1.39	9	6.75	4.03	-3.74	2.89	.01
Age	11	8.33	2.12	9	11.13	2.95	-2.80	2.39	.05

*One-tailed t test

Note: None of the WISC subtests showed significant differences between hyperactive and nonhyperactive patients.

TABLE 4

Differences Between Hyperactive Patients and Their Controls and Between Nonhyperactive Patients and Their Controls

TEST	N (PAIRS)	MEAN SCORE (PATIENTS)	MEAN SCORE (CONTROLS)	MEAN DIFFERENCE	S.D.	t	p<
Wechsler Intelligence Test for Children:							
Full-Scale IQ:							
Hyperactive	11	105.73	112.82	- 7.09	9.15	2.57	.02
Nonhyperactive	8	115.00	108.38	6.62	10.32	1.82	.05*
Verbal IQ (V):							
Hyperactive	11	108.00	110.91	- 2.91	13.13	0.73	NS
Nonhyperactive	8	117.50	115.00	2.50	14.30	0.49	NS
Performance IQ (P):							
Hyperactive	11	102.36	112.64	-10.28	16.10	2.12	.05
Nonhyperactive	8	109.25	99.63	9.62	11.67	2.33	.05
Difference V-P:							
Hyperactive	11	5.64	-1.73	7.37	24.23	1.01	NS
Nonhyperactive	8	8.25	15.38	- 7.13	18.26	1.10	NS
Draw-a-Person IQ:							
Hyperactive	10	90.00	93.70	- 3.70	28.91	0.40	NS
Nonhyperactive	8	94.75	84.00	10.75	13.48	2.26	.05
Bender-Gestalt (errors):							
Hyperactive	11	4.82	0.55	4.27	4.24	3.34	.01
Nonhyperactive	8	0.75	0.25	0.50	1.20	1.18	NS
Graham-Kendall (errors):							
Hyperactive	10	7.80	1.50	6.30	9.44	2.11	.05
Nonhyperactive	8	1.13	0.75	0.38	1.60	0.66	NS
Minnesota Percepto-Diagnostic (errors):							
Hyperactive	9	59.22	45.11	14.11	20.37	2.08	.05*
Nonhyperactive	8	44.38	34.50	9.88	14.97	1.87	.05*
Wide Range Achievement:							
Reading grade level:							
Hyperactive	11	3.37	4.55	- 1.18	1.86	2.11	.05
Nonhyperactive	9	8.34	6.65	1.69	2.78	1.83	.05*
Spelling grade level:							
Hyperactive	11	2.81	3.60	- 0.79	1.41	1.86	.05*
Nonhyperactive	9	6.58	6.02	0.56	2.60	0.64	NS
Arithmetic grade level:							
Hyperactive	11	3.01	3.13	- 0.12	0.64	.61	NS
Nonhyperactive	9	6.76	5.37	1.39	2.22	1.88	.05*

*One-tailed t test

metry of arm elevation in the Romberg position, pendular knee jerks, fine tremor of outstretched hands, imperfect left-right orientation, impaired two-point discrimination, eyelid fluttering in Romberg position (eyes closed), dysgraphesthesia, facial nerve motor inequality, tendon reflex asymmetry, articulation defect, mild adiposogenital or other dysplasias (all more common in the patient children), and mixed or variable handedness, eye muscle imbalance, equivocal Romberg sign (eyes closed), and general hyporeflexia (all more common in the control children).

Electroencephalograms

The findings with regard to the EEG slow activity index (mean total slow percent [TS%]) are shown in table 7 and figure 1. Table 7 A shows that TS% was significantly greater in the patient group than in the control group as a whole (N =24 pairs). That this difference was due largely to the excessive amount of slow activity in the nonhyperactive patient subgroup is revealed by the data indicating that, although TS% was only slightly greater in this subgroup than in the hyperactive patient subgroup (difference not significant), TS% of the nonhyperactive patients was significantly greater than that of their controls. TS% of the hyperactive patients was slightly greater than that of their controls, but the difference was not significant. The significant difference between the TS% of the nonhyperactive patients and their controls as estimated by the t test (table 7A) was confirmed by analysis of covariance. Thus, when TS% was corrected for age variation in the two patient subgroups combined and their respective

TABLE 5
Total Neurological Soft Signs

COMPARISON GROUPS	N	TOTAL SOFT SIGNS	WILCOXON T	MANN-WHITNEY U	p<
All patients (P)	24	92			
All controls (C)	24	39			
Statistic			2.5		.01
Hyperactive P	11	43			
Nonhyperactive P	9	33			
Statistic				46.0	NS
C (for hyperactive P)	11	21			
C (for nonhyperactive P)	9	14			
Statistic				41.5	NS
Hyperactive P	11	43			
C (for hyperactive P)	11	21			
Statistic			2.0		.01
Nonhyperactive P	9	33			
C (for nonhyperactive P)	9	14			
Statistic			0.0		.01

controls combined, a significant difference in $\overline{TS}$% was found (N = 40; df = 3, 35; F = 2.87, p <.05). Further analysis of covariance on the nonhyperactive patients and their controls revealed that the significant difference in $\overline{TS}$% between the whole patient group and the whole control group was due to the difference between these two subgroups (N = 18; df = 1, 15; F = 13.80; p < .01). No such difference was found by similar analysis of covariance on the hyperactive patients and their controls (N = 22, df = 1, 19; F = 0.72, NS).

Some possibly important aspects of these comparisons are revealed in figure 1 and table 7B. When the points representing the plots of mean age and $\overline{TS}$% for the hyperactive patient subgroup and the nonhyperactive patient subgroup are connected, the slope of this line, as estimated by applying the analysis of covariance to the means, is found to be significantly positive (N = 20; df = 1, 18; F = 4.63, p <.05). In contrast, the slope of the line connecting the corresponding points for the two control subgroups is found, by the same method, to be significantly negative (N = 20; df = 1, 18; F = 9.27; p <.01). Relevant to this difference in the signs of the two slopes are the data on mean age x $\overline{TS}$% product-moment correlation coefficients in table 7B. It should be noted that, although each of the two patient subgroups was closely matched with its control subgroup for age (differences not significant; see figure 1), the mean age of the hyperactive patients was significantly less than that of the nonhyperactive patients (mean difference = 35.2 months, t = 2.62, p < .02), and likewise for the two control subgroups (mean difference = 33.7 months; t = 2.37; p < 0.5).

Since normally one would expect slow activity in the EEG to decline with age (23, figure 31), the negative slope of the line for

TABLE 6
Neurological Soft Signs in Patients and Matched Controls
(Sign test for two related samples)

	N (PAIRS)	SUM P	SUM C	X (NO. OF FEWER SIGNS)	N (NO. OF PAIRS SHOWING DIFFERENCES)	p< (1-TAILED)	p< (2-TAILED)
Equivocal Babinski, etc.	24	10	0	0	10	.001	.002
Dysdiadochokinesis	24	6	0	0	6	.016	.032
Lateral nystagmoid jerks	24	5	0	0	5	.031	.062
Unilateral winking defect	24	5	0	0	5	.031	.062
Overflow movements*	24	6	14	4	16	.038	.076
Disturbances of gait, and/or coordination, and/or muscle tone	24	15	5	1	12	.003	.006

*Note more controls than patients had overflow movements (adventitious movements of fingers and/or opening and closing eyes in Romberg position, eyes closed, upper extremities outstretched forward).

TABLE 7

Total Mean EEG Slow (4.0-7.9 c.p.s.) Activity ($\overline{TS}$ %)

A. DIFFERENCES IN $\overline{TS}$ %

GROUP	N	MEAN SCORE	GROUP	N	MEAN SCORE	MEAN DIFFERENCE	S.E.	t	p<
All P	24	8.93	All C	24	2.96	5.97	1.30	4.58	.001
HP	11	8.07	C(HP)	11	5.15	2.91	2.64	1.10	NS
N-HP	9	10.23	C(N-HP)	9	1.34	8.89	2.62	3.40	.01
HP	11	8.07	N-HP	9	10.23	-2.16	3.76	0.57	NS
C(HP)	11	5.15	C(N-HP)	9	1.34	3.81	2.69	1.42	NS

B. AGE X $\overline{TS}$ % CORRELATION COEFFICIENTS

GROUP	N	x (AGE) (MONTHS)	y($\overline{TS}$ %)	r (xy)	p<
All P	24	118.08	8.93	-0.25	NS
All C	24	115.25	2.06	-0.41	.05
HP + N-HP	20	117.7	9.04	-0.20	NS
C(HP) + C(N-HP)	20	115.1	3.44	-0.43	.05*
HP	11	102.2	8.07	-0.12	NS
N-HP	9	137.4	10.23	-0.57	NS**
C(HP)	11	99.9	5.15	-0.50	NS***
C(N-HP)	9	133.6	1.34	-0.33	NS

*one-tailed (if r = 0.44, p<.05 [two-tailed]) for df = 18.
**if r = 0.58, p< .05 (one-tailed) for df = 7.
***if r = 0.52, p< .05 (one-tailed) for df = 9.
Abbreviations: P: patients; HP: hyperactive patients; N-HP: nonhyperactive patients; C: controls; C (HP): controls for hyperactive patients; C (N-HP): controls for nonhyperactive patients.

the control subgroups is not unexpected, but the positive slope of the line for the patient groups indicates that some factor other than age differentiated the hyperactive from the nonhyperactive patients as regards $\overline{TS}$%. The nature of this putative factor is suggested by comparisons of the subgroup $\overline{TS}$% values in table 7A and the subgroup correlation coefficients in table 7B. Thus there are no significant differences in $\overline{TS}$% between the hyperactive patient subgroup and either its control subgroup or the nonhyperactive patient subgroup, but the tendency of $\overline{TS}$% to decline with increasing age (r values) is much less strong in the hyperactive patient subgroup than in either of the other two, which are very comparable in this respect (the smaller r in the control subgroup for the nonhyperactive patients is to be expected since $\overline{TS}$% for this subgroup is very small). This suggests that whereas, in respect of $\overline{TS}$%, the two control subgroups may be regarded as homogeneous except for mean age differences, the two patient subgroups differ not only in regard to mean ages but also in regard to the origins of slow activity in their electroencephalograms. Inasmuch as for the patient group as a whole, the Spearman's Rank Correlation Coefficient for $\overline{TS}$% x number of soft signs was nonsignificant (r_S = 0.03), it may be presumed that the origins of slow activity in the electroencephalogram and of neurological soft signs are different (in the control group as a whole, the correlation coefficient for $\overline{TS}$% x number of soft signs, r_S = -0.07, was likewise

FIGURE 1

Mean total slow (4.0-7.9 c.p.s.) rhythmic EEG activity per channel ($\overline{TS}$%, ordinate) as a function of mean age (months, abscissa) in children with scholastic-behavioral problems and matched controls.

The points plotted are subgroup means (with standard deviations). A, definitely hyperactive patients (N=11); C, their matched controls (N=11). B, definitely nonhyperactive patients (N=9); D, their matched controls (N=9). Line A-B connecting the two points for the patient subgroups is given by the equation y=0.625x + 1.6827 (slope significant and positive in sign); line C-D connecting the two points for the control subgroups is given by the equation y=-0.1133x + 16.4720 (slope significant and negative in sign). For discussion, see text.

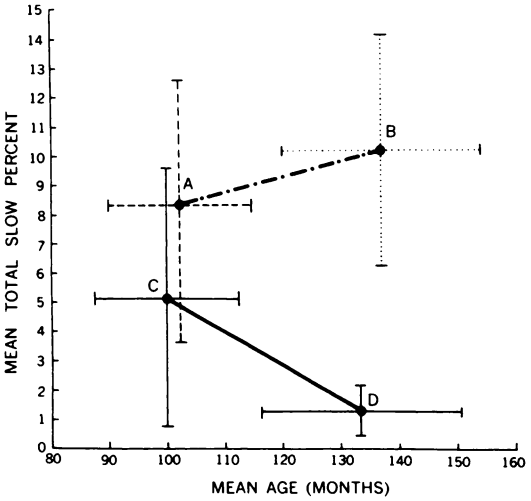


TABLE 8
Abnormal EEG Transients

A. ALL PATIENTS (N=24) VS. ALL CONTROLS (N=24)	SUM P	SUM C	X (NUMBER OF FEWER SIGNS)	N (NUMBER OF PAIRS SHOWING DIFFERENCES)	p< (ONE-TAILED)	p< (TWO-TAILED)
Abnormalities present	14	1	0	13	.001	.002

B. HYPERACTIVE PATIENTS (N=11) VS. NONHYPERACTIVE PATIENTS (N=9)	PRESENT	ABSENT	TOTAL
Hyperactive patients	8	3	11
Nonhyperactive patients	4	5	9
Total	12	8	20

Note: df = 1; chi square = 0.68, NS.

nonsignificant).

Abnormal EEG transients were rare in the waking records of both the patients and controls but fairly common during drowsiness or sleep in the patient group. The transients found in the patient group included: multifocal negative spikes in two, bitemporal repetitive 6 c.p.s. spikes in one, 8 c.p.s. spikes in one and 9 c.p.s. spikes in one, random spikes in one, focal spikes in one, paroxysmal spike-wave discharges in three, sharp waves in four, paroxysmal high-voltage theta discharges in 11, and focal theta in three. The electroencephalograms of the controls were free from such transient abnormalities except for the occurrence of diffuse paroxysmal theta discharges (during sleep) in one subject. During hyperventilation marked generalized slowing of the electroencephalograms was observed in 15 of the patients and in 13 of the controls; after resumption of normal breathing the duration of these varied between ten and more than 140 seconds in the patients and ten to 70 seconds in the controls. The presence of abnormal EEG transients was significantly greater in the patient group as a whole than in the control group as a whole (table 8A), but the hyperactive and nonhyperactive subgroups did not differ significantly in this regard (table 8B).

Discussion

Because of the heterogeneity of the population samples studied and the measures employed by different investigators, it is difficult to compare our results directly with those in the literature. In general, however, our overall findings are in agreement with those of others who have reported high incidences of perceptual-motor deficits (11, 17-19), neurological soft signs (14-17), and EEG abnormalities (4-13) in children with behavior problems. In this connection, we can concur in the comment of Hertzig and associates (16): "The findings ... make it difficult to sustain the view that the designation of the 'brain-damaged' is merely an 'administrative wastebasket' for poor learners or children with behavior problems. Clearly . . . the overwhelming majority of such children do indeed have clinical evidence of central nervous system abnormality."

Emerging from our study, however, are two distinct syndromes: one that is characterized by hyperactivity, perceptual-motor deficits, excessive neurological soft signs, and, in the EEG, excessive non-age-dependent slow activity and abnormal transient discharges; and another that is characterized by excessive neurological soft signs and, in the electroencephalogram, excessive but age-dependent slow activity and abnormal transient discharges without hyperactivity or perceptual-motor deficits. Inasmuch as our hyperactive subgroup was significantly younger than our nonhyperactive subgroup, and as hyperactivity tends to disappear as the child grows older (24), the possibility exists that the presumed central nervous system abnormality underlying the two syndromes is actually the same, with different rates of disappearance of its clinical manifestations over time—hyperactivity and perceptual-motor deficits receding earlier, and excessive EEG slow activity considerably later (cf. table 6B, r (age x $\overline{TS}$%) = −0.57 for the nonhyperactive patients).

This might be the case if, in our cross-sectional (not longitudinal) study, a selection factor had operated so that the nonhyperactive patient group represented children with more severe brain damage who,

had they been brought to the outpatient department when they were younger, would have been classified as hyperactive and found to have perceptual-motor deficits and very excessive amounts of slow activity in the EEG.[1] With increase in age, up to the time they were admitted to our study, the hyperactivity and perceptual-motor deficits might have disappeared, with, however, persistence of excessive EEG slow activity, thus accounting for the positive slope of the line connecting the TS% for the hyperactive and nonhyperactive subgroups in figure 1.

However, in the absence of evidence that such a selection factor did operate, it is also possible that the putative central nervous system abnormality underlying the two syndromes is not the same. Support for this dualistic hypothesis is furnished by the data of Laufer and associates (6), who found that whereas the photometrazol threshold for evocation of EEG spike-dome discharges and myoclonic jerks was abnormally low in hyperactive children, it was normal in nonhyperactive children with behavior disorders. Also, like ourselves, Klinkerfuss and associates (13) noted that the abnormally slow EEG activity of hyperactive children did not increase or decrease as age increased (nonhyperactive children were not studied). Resolution of the choice between the unitary and dualistic hypotheses must await the outcome of longitudinal studies on patient children such as those represented by our hyperactive subgroup.

Summary

Psychometric, neurological, and electroencephalographic studies were made of 24 children from five to 15 years of age who were brought to the psychiatric outpatient clinic because of a variety of scholastic-behavioral problems but showed no classical evidence of neurological disease. Twenty-four public school children matched with the patients for age, race, sex, I.Q., and socioeconomic class, who were doing well scholastically, had no record of behavioral problems, and were not known to have had neurological disease were also studied.

As a whole, the patient group differed significantly from the control group by having: lower scores on psychometric tests involving perceptual-motor tasks; a greater number of neurological soft signs; and greater incidence of diffuse, rhythmic slow (4.0-7.9 c.p.s.) activity, and greater incidence of abnormal transient discharges (spikes, paroxysmal discharges, etc.), mainly during drowsiness or sleep, in their EEGs.

Comparisons of the 11 definitely hyperactive and the nine definitely nonhyperactive patient children, both with each other and with their respective controls, revealed two distinct syndromes. One was characterized by hyperactivity, perceptual-motor deficits, an excess number of neurological soft signs and, in the EEG, excessive non-age-dependent slow activity and abnormal transient discharges. The other was characterized by an excessive number of neurological soft signs and, in the EEG, excessive but age-dependent slow activity and abnormal transient discharges, without hyperactivity or perceptual-motor deficits.

It is concluded that while in general, children with scholastic-behavioral problems do show evidence of brain dysfunction, the presumptive underlying central nervous system abnormality in the hyperactive children differs from that in the nonhyperactive children either in kind or in severity. Longitudinal studies on hyperactive children are needed to resolve this question.

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[1] The possibility of such a selection factor operating in the admission of the "nonhyperactive" patient children to this study was pointed out by Richard L. Jenkins, M.D., chief, child psychiatric service, University of Iowa State Psychopathic Hospital, Iowa City, Iowa.

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REFERENCES

1. Strauss AA, Kephart NC: Behavior differences in mentally retarded children measured by a new behavior rating scale. Amer J Psychiat 96:1117-1124, 1940
2. Strauss AA, Werner H: The mental organization of the brain-injured mentally defective child. Amer J Psychiat 97:1194-1203, 1941
3. Clements SD: Minimal Brain Dysfunction in Children. Terminology and Identification. Phase One of a Three-Phase Project, Public Health Service publication no. 1415. National Institute of Neurological Diseases and Blindness Monograph No. 3. Washington, DC, US Government Printing Office, 1966
4. Hill D: EEG in episodic psychotic and psychopathic behaviour. Electroenceph Clin Neurophysiol 4: 419-442, 1952
5. Forsmann H, Frey TS: Electroencephalograms of boys with behavior disorders. Acta Psychiat Neurol Scand 28: 61-73, 1953
6. Laufer MW, Denhoff E, Solomons G: Hyperkinetic impulse disorder in children's behavior problems. Psychosom Med 19: 38-49, 1957
7. Daveau M: The EEG of 150 children with behavior disorders. Electroenceph Clin Neurophysiol 10: 198, 1958
8. Kelloway P, Crawley JW, Kagawa N: A specific electroencephalographic correlate of convulsive equivalent disorders in children. J Pediat 55:582-592, 1959
9. Kennard MA: The characteristics of thought disturbances as related to electroencephalographic findings in children and adolescents. Amer J Psychiat 115: 911-921, 1959
10. Green JB: Association of behavior disorder with an electroencephalographic focus in children without seizure. Neurology 11: 337-344, 1961
11. Anderson WW: The hyperkinetic child: a neurological appraisal. Neurology 13: 968-973, 1963
12. Gross MD, Wilson WC: Behavior disorders of children with cerebral dysrhythmias. Arch Gen Psychiat 11: 610-619, 1964
13. Klinkerfuss GH, Lange PH, Weinberg WA, et al: Electroencephalographic abnormalities of children with hyperkinetic behavior. Neurology 15: 883-891, 1965
14. Kennard MA: Value of equivocal signs in neurologic diagnosis. Neurology 10: 753-764, 1960
15. Larsen VL: Physical characteristics of disturbed adolescents. Arch Gen Psychiat 10:55-88, 1964
16. Hertzig ME, Bortner M, Birch HG: Neurologic findings in children educationally designated as "brain-damaged." Amer J Orthopsychiat 39: 437-446, 1969
17. Clements SD, Peters JE: Minimal brain dysfunction in the school age child. Arch Gen Psychiat 6: 185-187, 1963
18. Diller L, Birch HG: Psychological evaluation of children with cerebral damage, in Brain Damage in Children. The Biological and Social Aspects. Edited by Birch HG. Baltimore, Williams and Wilkins, 1964, pp 27-45
19. Stevens DA, Boydstun JA, Dykman RA, et al: Presumed minimal brain dysfunction in children. Arch Gen Psychiat 16: 281-285, 1967
20. Edwards AL: Statistical Methods, 2nd ed. New York, Holt, Rinehart and Winston, 1967, pp 97-121 and 200-225
21. Edwards AL: Experimental Design in Psychological Research, 3rd ed. New York, Holt, Rinehart and Winston, 1968, pp 326-347
22. Siegel S: Nonparametric Statistics. New York, McGraw-Hill, 1956, pp 68-75, 75-83, 104-111, 116-127 and 202-213
23. Gibbs FA, Gibbs EL: Atlas of Electroencephalography, vol 1, 2nd ed. Reading, Mass, Addison-Wesley, 1951, p 84
24. Eisenberg L: Behavioral manifestations of cerebral damage in childhood, in Brain Damage in Children. The Biological and Social Aspects. Edited by Birch HG. Baltimore, Williams and Wilkins, 1964, pp 61-76